Liam Leirs

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LinkedIn
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Portfolio

Profile

Master of Science in Data Science and Artificial Intelligence from the University of Antwerp, graduated with distinction, with a strong background in computer science. Experienced in machine learning, deep learning, recommender systems, computer vision, and reinforcement learning through academic research and personal projects. Interested in developing and evaluating data-driven systems and applying AI to real-world problems.

Education

Master of Science in Data Science and Artificial Intelligence 2024 – 2026
University of Antwerp – *Graduated with Distinction*

Bachelor of Science in Computer Science 2021 – 2024
University of Antwerp – *Graduated with Distinction*

Secondary Education – Mathematics-Science 2015 – 2021
Sint-Michielscollege Schoten

Technical Skills

Programming & Data

- Python, C++, SQL, NumPy, Pandas

Machine Learning & AI

- PyTorch, deep learning, model training, hyperparameter optimization, and evaluation
- Recommender systems, computer vision, reinforcement learning, explainable AI

Data Science

- Data preprocessing, exploratory data analysis, statistical analysis, and visualization

Tools

- Git, LaTeX, Jupyter

Projects

Master Thesis – GRU4Rec Reproducibility 2025 – 2026

- Investigated performance differences between RecPack and the official GRU4Rec implementation for session-based recommendation
- Performed code-level analysis, controlled experiments, and independent hyperparameter optimization to identify sources of performance differences
- Demonstrated the impact of implementation and experimental choices on recommender-system benchmarking and reproducibility

Research Project – Network Pruning for WSOL 2025 – 2026

- Investigated the effect of neural network pruning on Weakly Supervised Object Localization using ResNet18 and VGG16
- Evaluated magnitude, Filter-Norm, and FPGM pruning using Grad-CAM, Layer-CAM, and Score-CAM

- Found that pruning can improve object localization despite decreasing classification accuracy, particularly in overparameterized networks

Reinforcement Learning – Connect Four Self-Play 2026

- Developed a custom Connect Four reinforcement learning environment and trained a Maskable PPO agent through self-play
- Implemented opponent leagues, Elo-based evaluation, model checkpoints, action masking, and varied starting states
- Benchmarked learned policies against random, tactical, MiniMax, and previous PPO agents

Experience

Kitchen & Logistics Assistant – Hof van Schoten 2018 – 2026

- Supported daily kitchen and logistics operations in a residential care environment
- Collaborated with multidisciplinary staff while working in a structured and safety-conscious environment

Languages

Dutch (Native)
English (Fluent)